GROUP 39 INDIVIDUIAL REPORTS

https://github.com/rosieguiney/COM109-Coursework2

Group 39 Individual Report – Lily Lockhart, B01016864

INTRODUCTION

For our coursework 2 as a group we decided on a mindfulness website that aims to reduce stress and anxiety around schoolwork which is where we came up with the name Stress Less. Stress is something that everyone deals with so we thought we could offer a site with a range of services to help reduce these anxieties. Our main focus for this project was to provide a good user interface and user experience along with consistency and accessibility such as including alt tags for photos. We were trying to implement as many of Shneiderman's eight golden rules as possible because these are the crucial foundations for providing a user-friendly website.

REQUIREMENTS

In our first meeting we researched likeminded websites and came across mind.org where we took creativity from it in regard to what pages we should include and what features we thought looked good such as a "back to top" button and a virtual assistant chat box. With this in mind we decided our website should consist of five pages: Home, about us, coping tools, resources and a contact page. Other requirements we thought would be good to implement was providing our users with a chance to give feedback allowing user interaction and a chance for them to suggest improvements for our website. Other requirements we thought could be useful were a hamburger menu, a navigation bar at the header as this makes it easier for the user to navigate between pages and button hover effects. Since we wanted to provide a calm, peaceful environment it was decided that pastel colour scheme would be best way to implement this, so we decided on #FFE7FD as our background colour and a gradient of this for the breathing techniques. We also used Miro as our collaborative tool to store all ideas so when it came to implementing them, we could refer back. In our second meeting we decided it would be best to split the workload into pages and as a result, I was assigned the home page. A wireframe was created on Canva to help provide a visual aid and what the main page should look like.

Canva wireframe:

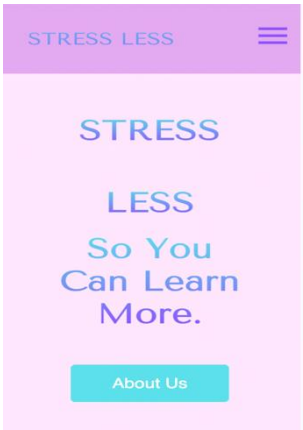
Colour scheme:

DESIGN

For the creation of the home page, I used a range of tools consisting of HTML5, JavaScript, CSS3 and jQuery along with Visual Studio Code and GitHub to combine everyone's work. Being tasked with the home page meant it was crucial everything looked appealing and pleasing to the eye which as a result would entice users to explore more of the website and see the services we have to offer. To attain this idea, I created an inviting hero section which was done by using a container. The container was formatted so it controlled how my text and image sat. This along with the background image selector allowed me to generate the gradient for the text creating a warm and inviting environment for our clients. I also provided an "about us" button below so if users had any inquiries at the start, they would hopefully be answered in that page aiding accessibility. A problem I ran into when developing this was when I went to split screen the text and image started to overlap. To combat this, I added media queries throughout the home page to change the layout to a single column on smaller screens so it would be more responsive.

```css
.main__container {
    display: grid;
    grid-template-columns: 1fr 1fr;
    align-items: center;
    justify-self: center;
    margin: 0 auto;
    height: 90vh;
    background-color: #FFE7FD;
    z-index: 1;
    width: 100%;
    max-width: 1300px;
    padding: 0 50px;
}
```

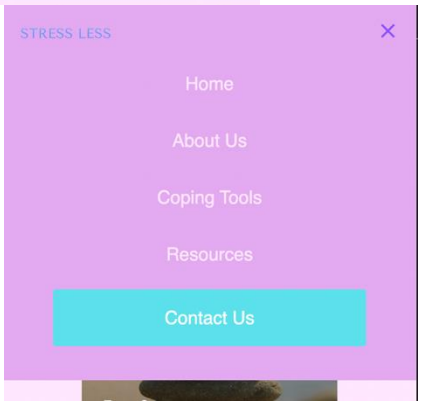
```css
@media screen and (max-width: 480px) {
    .main__content h1 {
        font-size: 2rem;
        margin-top: 3rem;
    }

    .main__content p {
        margin-top: 2rem;
        font-size: 1.5rem;
    }

    .main__btn {
        padding: 12px 36px;
        margin: 2.5rem 0;
    }
}
```



Media query in action

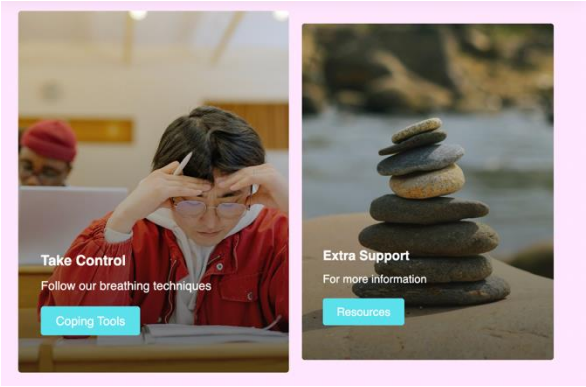
To aid navigation and accessibility in the website, I designed a fully functional header and footer which was shared on GitHub so everyone in the group could download and use further aiding the consistency of each page. The navigation bar provides full and easy navigation around the website. To make it fully functional and more responsive, when the user goes into split screen mode it automatically turns into a hamburger menu because likewise with above, the text started to become cramped and unappealing to the eye. JavaScript was used to implement this through an event listener which detected when the navigation menu icon was clicked. By doing this the user can now still easily move between the different pages. Another feature of the navigation bar is that when the user scrolls down the page it will still remain at the top further aiding functionality. With the footer it consists of the "most popular" links such as FAQs and social media links, further enhancing responsiveness and useability to the website.

```
const menu = document.querySelector('#mobile-menu');
const menuLinks = document.querySelector('.navbar__menu');

menu.addEventListener('click', function() {
    menu.classList.toggle('is-active');
    menuLinks.classList.toggle('active');
});
```



To show off what services we have to offer I thought it would be best to implement these using cards since it helps to structure the content better in terms of organising information clearly, improving readability and focus and helping to guide user action since there is a button at the bottom of each card to guide users to each service available. To make this section feel more modern I added hover effects over each card with the hope of improving engagement as it makes users feel like they are being enticed to click on the button and explore.

Example of hover:




```
.services__card:hover {
    transform: scale(1.075);
    transition: 0.2s ease-in;
    cursor: pointer;
}
```

When creating the buttons I made sure they followed the same structure to keep a consistent and professional feel to the page. To make this feature even more engaging, I decided to add hover effects that were implemented in CSS using the :hover class. I opted for a transition to make it smooth instead of instant because it draws the users' attention gradually. Using smooth animations also make the website feel more modern and of a higher quality.

Example of hover effect:

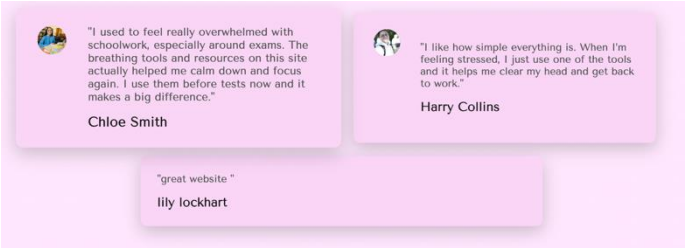
```
.services__card button:hover {
    background-color: rgba(60, 180, 200, 0.85);
    transition: background-color 0.3s ease;
}
```

Although the hamburger menu and cards were the main way in which I fulfilled my user requirements for the home page I also included other features which assisted me in improve functionality and accessibility. I decided to add a student testimonial section along with a “leave your own comment” feature where students could post comments to the page. To add new comments to the webpage I used a jQuery. By doing this it further aids user involvement due to the option of feedback providing a sense of community and therefore possibly leading to return use. Since it was created using the form element, I ensured that I used validation meaning the form could not be submitted without input. To further validate it even after the user hits post another check comes up confirming if they want to post it or not.

Pop up check:

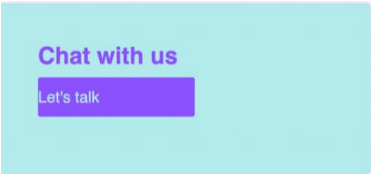
```
const confirmPost = confirm("Are you sure you want to post this comment?");

if (!confirmPost) return;

const newComment = `
    <div class="testimonial-col" style="display:none;">
        <div>
            <p>"${comment}"</p>
            <h3>${name}</h3>
        </div>
    </div>
`;

const element = $(newComment);
$('.user-comments').append(element);
element.fadeIn(500);

this.reset();

alert("Thank you for the feedback!");
```

Comment added:

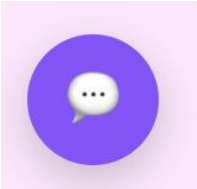


Another interactive feature that was a good addition to the website was the virtual chat box that Rosie created for the contact page. With the aim of a user-friendly system in mind we decided to implement this feature on all of the pages because it provides them with constant access to help and with this reduces user frustration. Since our website is about reducing stress, it fits with the image and makes it more effective. A feature that I changed to the chat box was when the user clicks on the button, it opens to the left instead of the right. This was so it doesn't interfere and block the "back to top" button that Joe created and enhances user experience by making the feature constantly accessible. However, a problem with this chat box is that it may not be clear that it's a help chat box and so as a result this helpful feature may go unrecognised.

Chat box in contact page:

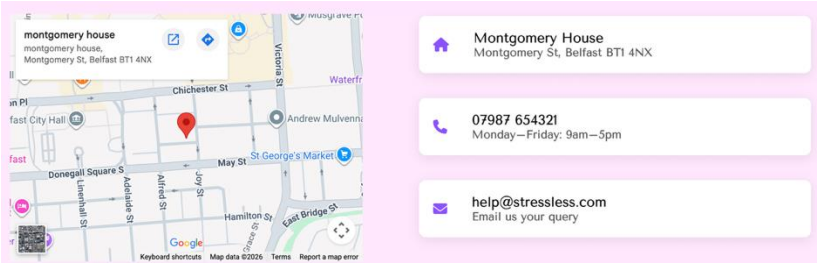


chat box in other pages:



```
/* CHAT BUTTON */
.chat-button {
    position: fixed;
    bottom: 20px;
    left: 20px;
    width: 55px;
    height: 55px;
    border-radius: 50%;
    border: none;
    background: #8C52FF;
    color: white;
    font-size: 22px;
    cursor: pointer;
    box-shadow: 0 6px 18px rgba(0,0,0,0.2);
    z-index: 1000;
    transition: all 0.3s ease;
}

.chat-button:hover {
    transform: scale(1.1);
    background: #6f3de0;
}
```

To further assist the idea of a user-friendly system, I adopted a "reach out" section to provide users with clear and accessible contact information. To attain this, I added an embedded map which shows the location and contact details which use a flexbox layout to organise the content side by side. I styled this section to make it visually distinct from the rest of the page which helps to draw attention and ensure users can easily find information. Likewise I also implemented media queries to allow the layout to adapt on smaller screens.

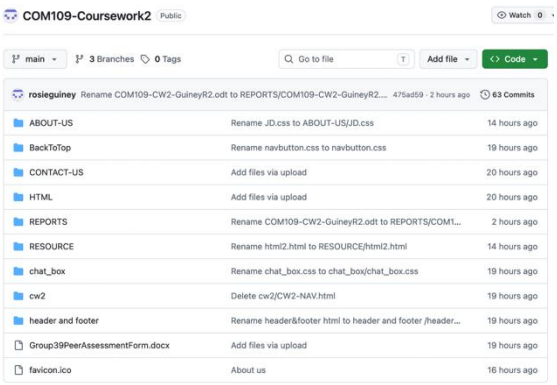


```css
.contact-tag {
    display: inline-block;
    font-size: 0.75rem;
    letter-spacing: 2px;
    text-transform: uppercase;
    padding: 6px 12px;
    border-radius: 50px;
    background: linear-gradient(90deg, #8C52FF, #5CE1EC);
    color: white;
    margin-bottom: 10px;
}
```

I created a quote box using JavaScript to display motivational messages to users. When the button is clicked, a random quote is selected from an array using a random number generator and displayed on the screen. This adds interactivity to the website and provides users with positive reinforcement, which supports the overall theme of stress reduction. The feature improves user engagement by allowing users to generate new messages whenever they choose. However, a problem I ran into with this is that since the quote box uses a fixed array of messages, users will see the same quotes repeated which can make the feature feel less engaging if clients are constantly using it. If I had more time to do this again, I would link the quotes to an external database or categorise them based on mood or context, so it doesn't become repetitive.

IMPLEMENTATION

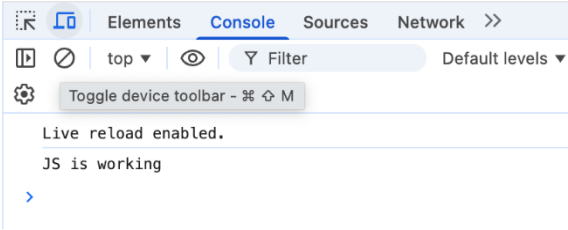
As a group we decided on the final date that we would have our webpages complete and uploaded this to GitHub. This allowed for ease of use as every member could see what was uploaded and when. It also made it easier for me to connect everything together in the navigation bar to make the website complete and seamless. Using GitHub meant that there was always an extra copy in case something got corrupt or it failed. The webpages were linked using the href element in the navigation bar.




```html
</div>
    <ul class="navbar__menu">
        <li class="navbar__item">
            <a href="index.html" class="navbar__links">Home</a>
        </li>
        <li class="navbar__item">
            <a href="AboutUs.html" class="navbar__links">About Us</a>
        </li>
        <li class="navbar__item">
            <a href="breathing.html" class="navbar__links">Coping Tools</a>
        </li>
        <li class="navbar__item">
            <a href="html2.html " class="navbar__links">Resources</a>
        </li>
        <li class="navbar__btn">
            <a href="contactHTML.html" class="button">Contact Us</a>
        </li>
    </ul>
```

TEST

To ensure my webpage worked as it should, I carried out a number of tests. The first one was to ensure that my JavaScript link was working in my html document. To carry out this test I used a console log to confirm it appeared in the console and so consequent JavaScript would work.



The second test was to ensure that in my “leave a comment” feature, I couldn’t submit it with the fields blank. To achieve this a jQuery was used with this implementation for checking if the fields are empty and using DOM manipulation to add the new comments dynamically. It also makes sure the form is reset afterwords.

```jquery
$('.comment-form').on('submit', function(e) {
    e.preventDefault();

    const name = $('.comment-form input').val();
    const comment = $('.comment-form textarea').val();

    if (!name || !comment) {
        alert("Please fill in both fields!");
        return;
    }
```

Another test that I carried out was to make sure that the navigation links actually worked in the header like they were supposed to. To test this I simply clicked on every link and made sure it took me to the correct place on the webpage.

OVERVIEW

Overall, we worked well as a group with constant communication and feedback with each other on WhatsApp and GitHub. Over coursework 2 we were able to create an interactive and user-friendly system that could obtain user feedback and met our requirements outlined at the start. If I was to complete this project again, I would consider implementing different features such as a dark mode or even interactive map improvements such as allowing users to search for locations or get directions.

Reference list

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Easy Tutorials (2021). *How To Make Website Using HTML & CSS | Full Responsive Multi Page Website Design Step by Step*. *YouTube*. Available at: https://www.youtube.com/watch?v=oYRda7UtuhA [Accessed Apr. 2026].

Krukau, Y. (2021). *Woman in Blue Long Sleeve Shirt Sitting on Chair · Free Stock Photo*. [online] Pexels. Available at: https://www.pexels.com/photo/woman-in-blue-long-sleeve-shirt-sitting-on-chair-8199559/ [Accessed Apr. 2026].

Krukau, Y. (2022). *Man in Red Jacket Studying at a Classroom · Free Stock Photo*. [online] Pexels. Available at: https://www.pexels.com/photo/man-in-red-jacket-studying-at-a-classroom-8197500/ [Accessed Apr. 2026].

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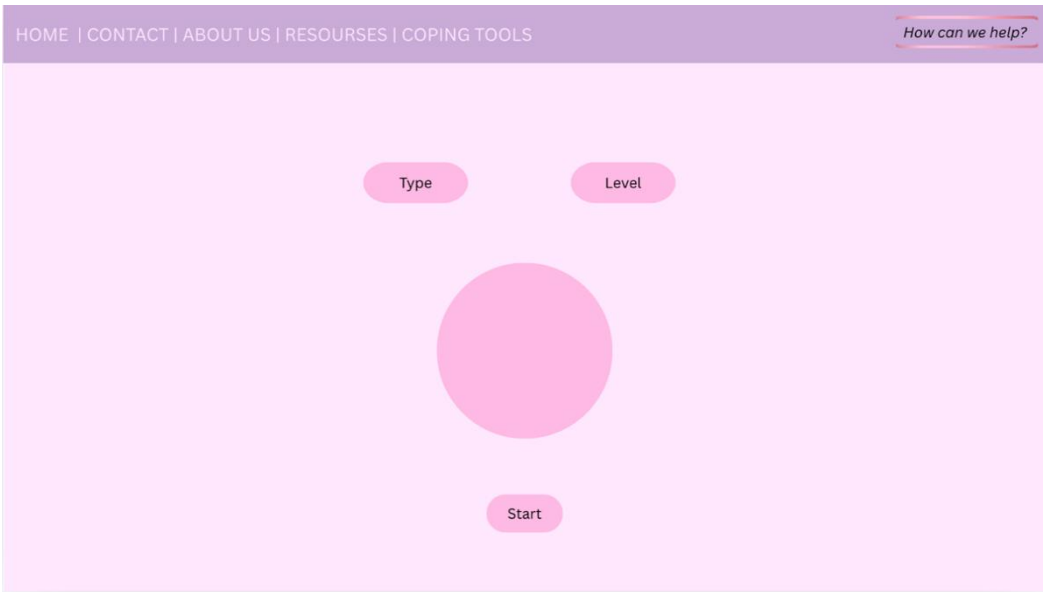
Group 39 Individual Report- Siân McManus-B01021399

Intro

This group project involved the collaborative design and development of a web-based application which we named '*Stress Less'*, that aims to help students manage and reduce stress. The website was developed using a mix of HTML, CSS, JavaScript and jQuery. With a strong focus on interactivity, usability and accessibility. As a group, we planned, designed and implemented the system collaboratively. Throughout the development process, we aimed to apply **Shneiderman's Eight Golden Rules of Interface Design** particularly with regards to consistency and usability to ensure a clear and user-friendly experience.

Throughout the project, we organised several group meetings to plan the structure and overall functionality of our website. During these meetings we discussed who our target audience would be and identified key features that we wanted and considered how to create an effective and user-friendly experience for our website.

The use of Miro allowed the group to visually map out ideas and how the features would all work together, which improved our communication and reduced misunderstandings during the development. Similarly, Canva was used to produce initial wireframes and logo concepts, helping to form a clear visual direction before implementation began. This planning stage ensured that development was more structured and efficient.



(Initial wireframe)

To manage the workload we divided the website into different sections rather than working on everything together. This allowed each team member to take responsibility for a specific part and develop it into more detail. The breathing exercise section acts as an interactive feature of the website. This section allows users to select a breathing method and difficulty level and then it guides them through a structured breathing exercise using animations and timed instructions. The main aim of this section was to create a functional, visually appealing and relaxing webpage that was easy to use for all.

Design and Implementation

The breathing exercise is the main focus of my assigned webpage(breathing tools) and it makes use of asynchronous JavaScript through setTimeout() and setInterval(). These functions allow for controlled time-based events such as transitioning between inhale, hold, and exhale phases. By updating both the visual animation and countdown timer the webpage provides a real-time response to the user, improving usability and effectively guiding them through the breathing process. This reflects some principles of event-driven programming because as users interact (clicking the start button) it triggers a sequence of timed events.

I initially watched lots of tutorials on YouTube and read through lots of tutorials online to figure out how to make the breathing animations. I found a tutorial online that I understood and used it as a basis for this feature. I found W3schools particularly helpful when it came to finishing touches or core things that I was able to then build on. However, the implementation was significantly adapted to suit our website. Modifications that I made include restructuring the timing logic and improving synchronisation between the animation and having a different animation for the box breathing.

The use of difficulty levels introduces scalability to the project as the duration of each breathing phase is modified using a multiplier. Meaning that it can be used to create flexible and reusable code.

```
//variables controling the cycles
let isRunning = false;
let timeout;
let interval;
let phaseIndex = 0;
```

These variables manage the execution and progression of the breathing cycle. Initially I found it difficult to synchronise the animation with the instructions and the timer. Early versions of my code often resulted in inconsistent behaviour, where the animation would not match the instructions properly or the timer would reset at the wrong time. This made the feature confusing to use and highlighted that my logic needed to be improved. To address this I restructured the animation logic by using predefined breathing patterns. Each pattern was stored as an array of objects, with each object containing the instruction such as “Inhale”, “Hold”, or “Exhale” and a time value.

A key design decision was the use of structured JavaScript objects to represent both breathing patterns and difficulty levels. The ‘levels’ object maps the users selected difficulty options to values which are then used to dynamically adjust the speed of the breathing cycle. Similarly, the patterns object defines each breathing technique as sequence of phases with each phase containing instruction text and a time duration.

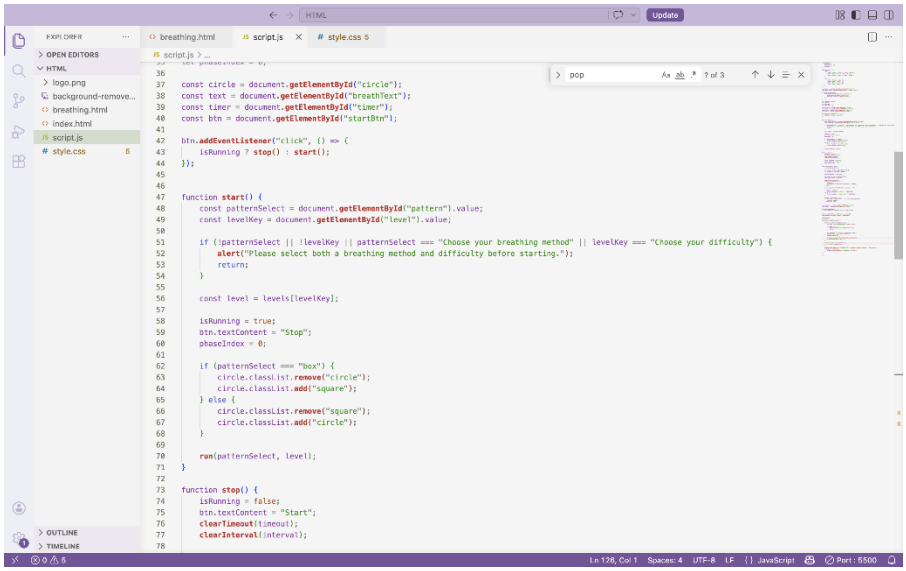
```
JS script.js > ...

//difficulty levels (higher is slower and harder)
const levels = {
    beginner: 2,
    intermediate: 1.5,
    advanced: 1
};

// diff breathing patterns
const patterns = {
    "478": [
        { text: "Inhale", time: 4 , class: "inhale" },
        { text: "Hold", time: 7, class: "hold" },
        { text: "Exhale", time: 7, class: "exhale" }
    ],

    "box": [
        { text: "Inhale", time: 3 },
        { text: "Hold", time: 3 },
        { text: "Exhale", time: 3 },
        { text: "Hold", time: 3}
    ]
};
```

Another issue I encountered was with the start and stop functionality. At first, clicking the “Start” button multiple times caused multiple timers to run at once, which made the animation behave unpredictably. Similarly, the “Stop” button did not fully reset the animation, leaving some processes running in the background. I resolved this by introducing a Boolean variable to track whether the animation was currently active and by using clearTimeout() and clearInterval() to stop any running timers when needed.

This ensured that only one animation cycle could run at a time and that it was more stable overall.

```
//variables controling the cycles
let isRunning = false;
let timeout;
let interval;
let phaseIndex = 0;
```

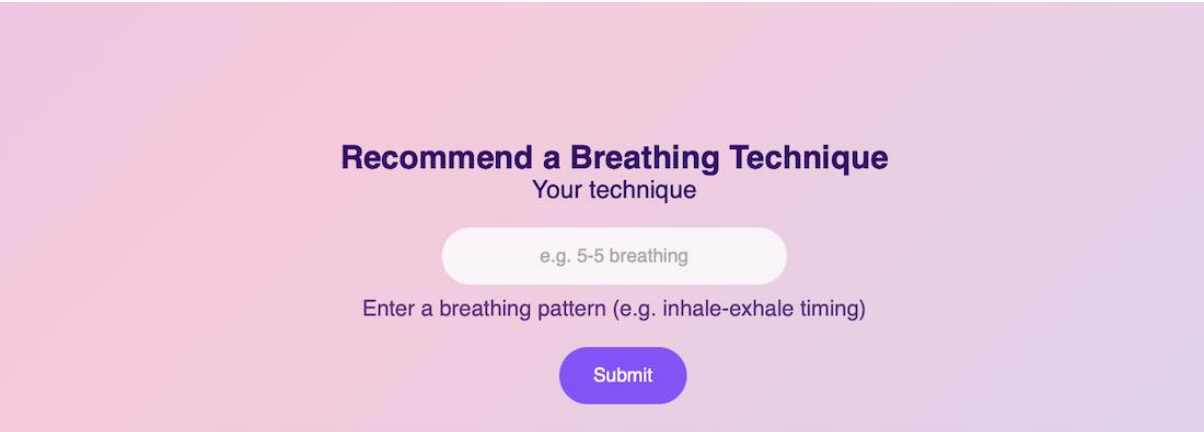
In addition to the animation, I also implemented validation within the breathing tools section. Users are required to select both a breathing method and a difficulty level before starting the exercise. In earlier versions, users could continue without making these selections which caused errors in the program. To fix this I added validation checks that display clear error messages if required inputs are missing. These messages are updated dynamically giving immediate feedback and improving the overall usability of the feature.



I implemented hover effects on my buttons so that it is clear to the user what they are selecting, increasing user engagement.

```
select:hover {
transform: translateY(-2px);
}

#startBtn {
padding: 12px 30px;
border-radius: 25px;
border: none;
background: linear-gradient(135deg, #cdb4ff, #ffc8dd);
color: #3c096c;
font-weight: 600;
cursor: pointer;
transition: 0.3s;
}

#startBtn:hover {
transform: translateY(-2px) scale(1.05);
box-shadow: 0 8px 20px rgba(0,0,0,0.1);
}
```

I also developed a recommendation feature that allows users to submit their own breathing techniques. This involved dynamically adding new items to the page using jQuery. Initially the feature allowed for empty and duplicate submissions, which reduced its effectiveness. To improve this, I added validation to trim whitespace, check for empty inputs and prevent duplicate entries by comparing the new input with existing list items. The recommendation system demonstrates dynamic DOM manipulation using jQuery.



When a user submits a new breathing technique, the application validates the input, checks for duplicates, and then updates the interface in real time without refreshing the page. This improves user experience and reflects modern web development practices. The inclusion of visual feedback through animations and popups further enhances interactivity.

I would have preferred to have had more than one jQuery implemented, but I needed to balance my workload and have a good well-rounded webpage. If I was to work on this further or have more time this is definitely something I would have done. Having to work on other coursework and study for other exams meant that I couldn't devote all of my time to this page.

```
checkReady();
// recommendation section
$(document).ready(function() {

    $("#recommendBtn").click(function() {
        let input = $("#recommendInput").val().trim();

        // empty check
        if (input === "") {
            alert("Please enter a technique first!");
            return;
```

To make the interaction feel more engaging I added a popup confirmation message when a user submits a recommendation, providing clear feedback that their input has been received. These small details help improve the overall user experience and make the website feel more responsive.

From a visual perspective, CSS transitions and transformations were dynamically controlled through the JavaScript. As well as the circular animation a box breathing technique was implemented using a square that moves between positions. This was achieved using CSS transformations.

```
.pos-0 { transform: translate(-135px, -135px); }
.pos-1 { transform: translate(135px, -135px); }
.pos-2 { transform: translate(135px, 135px); }
.pos-3 { transform: translate(-135px, 135px); }
```

I also made the Favicon so that it was **visibly identifiable to users** instantly among multiple open browser tabs. It also acts as a reinforcement of our brand identity, acting as a miniature logo.

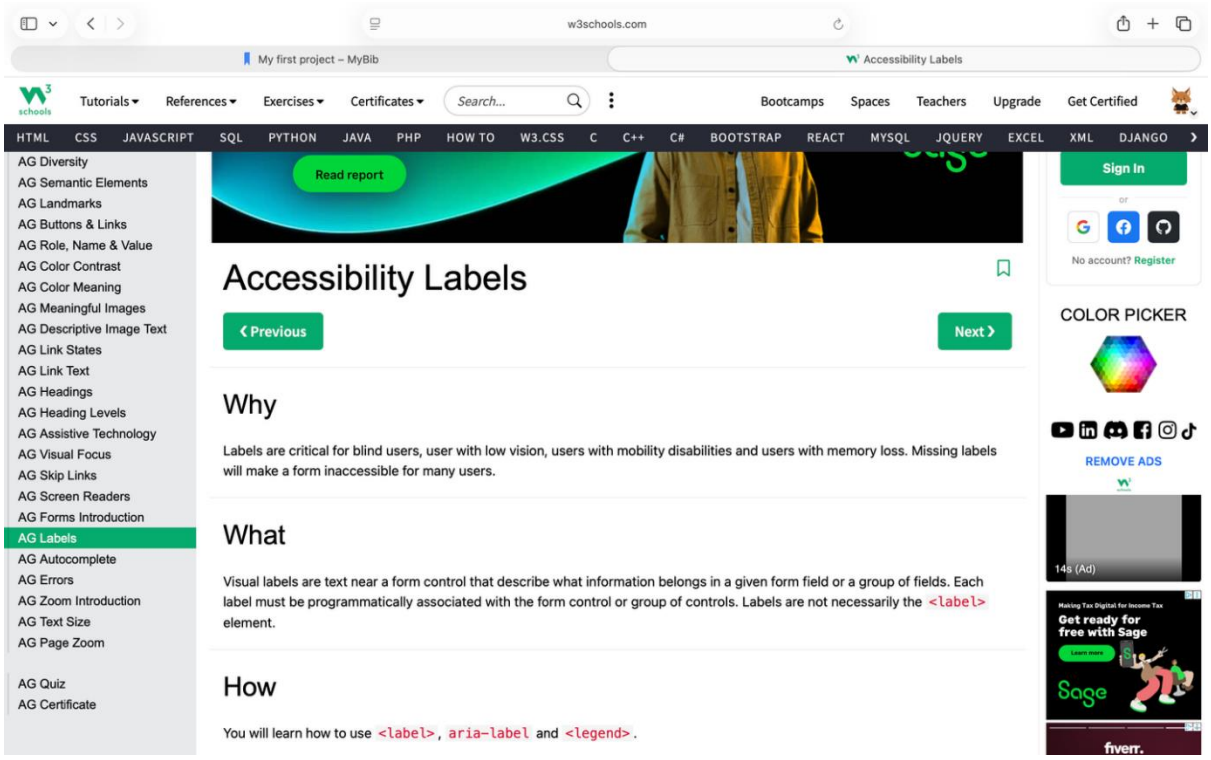
(Logo)

(Favicon)

From a design perspective, we intentionally aimed to develop the webpage with calm colours. We did this to keep the interface simple clear and calming. I used a soft gradient colour scheme with shades of purple, pink and blue to reflect the theme of relaxation, (135deg, #cdb4ff, #ffc8dd, #bde0fe). Research suggests that colours directly impact a person's emotions therefore a gentle palette was selected to align with the purpose of stress reduction The layout was designed to centre the main breathing tool so that it is easy for users to focus on the primary function of the site. Additionally, consistent font usage and rounded UI elements were implemented to create a modern interface. This design approach ensures that users feel comfortable and not overwhelmed when interacting with the application. However, achieving this layout was not as straightforward as I initially thought it would be. My initial use of absolute positioning caused alignment issues and overlapping elements. I later switched to using Flexbox, which allowed for better control over positioning and made the layout more consistent and easier to manage. This use of flexbox significantly improved my layout compared to my original use of absolute positioning.

Accessibility and Testing

Accessibility was also considered throughout the development process. I ensured that form inputs were linked with labels and included ARIA attributes such as aria-required and aria-live to support screen readers. I also made sure that users could interact with the site using the keyboard, for example by allowing the Enter key to submit input. These decisions help make the application more inclusive and accessible to a wider range of users. These considerations align with inclusive design principles ensuring that our website can be used by a wider range of users. While ARIA attributes improved accessibility, testing was not conducted with actual screen reader users, which limits my confidence in real-world usability. I used W3schools to help me implement the ARIA attributes.



Testing was carried out continuously during development. I tested different user inputs to ensure validation worked correctly and checked that the breathing animation behaved as expected under different conditions. I also focused on identifying timing issues, as these had a direct impact on the user experience. Testing was mainly carried out on desktop devices and changes were made based on any issues that were identified. However, there was limited mobile testing which may affect performance on smaller devices highlighting an area for future improvement.

Overall, this project was a valuable learning experience both technically and in terms of teamwork. It helped me develop a stronger understanding of JavaScript particularly when working with timing functions, DOM manipulation and event handling. It also highlighted the importance of planning, communication and dividing tasks effectively within a group. If I were to continue developing this project, I would focus on improving responsiveness for mobile devices, adding more advanced validation and implementing a way to store user input permanently. These improvements would further enhance both the functionality and user experience of the application. The interface was designed with usability in mind, ensuring that all key features are easily accessible. Clear labels, consistent styling, and immediate feedback help users understand how to interact with the system. However, further improvements could include accessibility enhancements such as more keyboard navigation. We used GitHub put everything together, although I did not understand how to use it during development I just

uploaded all of my work at the end instead of gradually adding things in as I developed it. When working in a group in the future I would learn how to use GitHub more effectively.

Reference list

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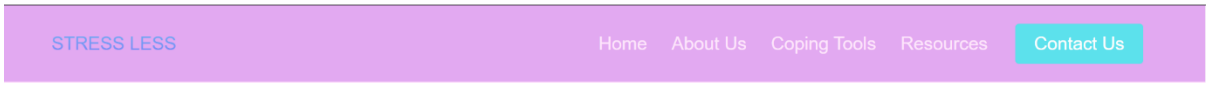
Rosie Guiney's Report

Introduction

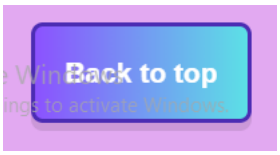
For this coursework our group choose to create a website revolving around mental health support, we thought this would work well for the coursework as we could meet all of the criteria and create a design without too much difficulty or it being too easy. My group consists of five members (Me, Lily, Siân, Joe & Malak) and during our first meeting we decided on some basic details for our website such as the name, logo, colour scheme and what pages we would like to include. We also created a WhatsApp group so we could communicate easier and created an area on GitHub to upload our code though none of us have used it before, so it was quite confusing. We used the app/website Canva to help design our logo and we used the website Miro to help plan everything else.

Before we came up with the basic design, we looked at similar websites to get an idea of what we should include on each page. We split the workload so that each of us would do one webpage each and we spun a random wheel to decide who would do each page to make it fair – After this we started the code. A few of us including myself designed our own navigation bar which probably wasn't necessary as we ended up just choosing one to keep our website consistent. During our coursework we had a 2-week break for easter so we ensured that we had everything we needed so that we could work individually on each page as it would have been challenging to meet up during this break. During this break we kept each other updated and once we came back we met up a few times to connect each webpage together and add final details.

Navigation Bar & Footer



The navigation bar and footer for our website were created by Lily. It is the same on every page to keep it consistent and it is very clear and accesible to users trying to navigate our website. The footer includes quick links that make it easy for users to find specific resources.



Joe created a back to top button which appears once you try scroll down on the website – we added this to make it easy for users to return back onto the top of the page. This would be useful if they wanted to reach the navigation bar to quickly change page.

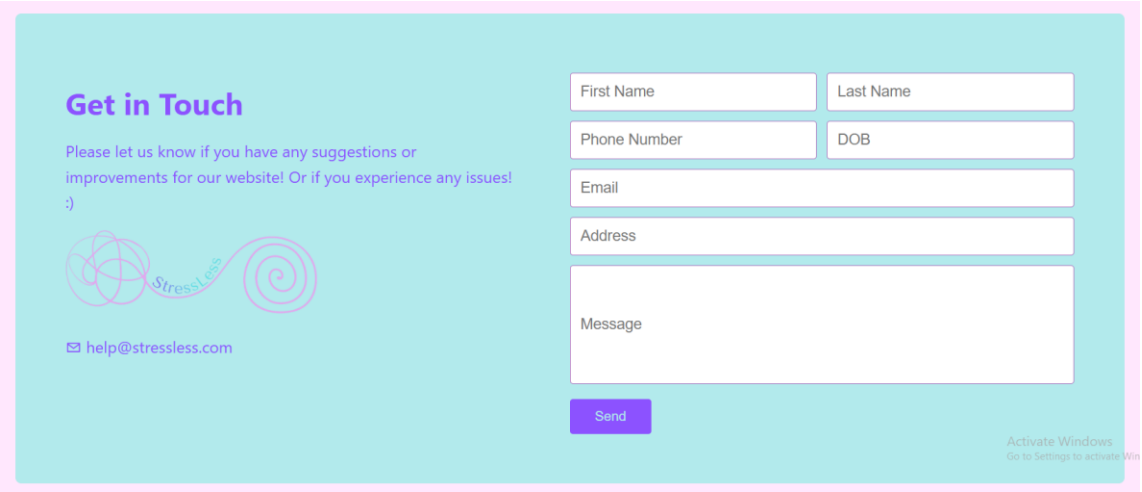
Contact page

This will be the largest section in this report as it was the page that I developed. I used HTML, CSS, JavaScript and jQuery to develop my webpage. I was happy to be able to create the contact page as I thought it would be able to implement it very well and I am happy with the outcome. My webpage has features such as animations, interactivity and form validation.

(Please ignore the “Activate Windows” in the corner of any screenshots it is part of my laptop screen and I cannot remove it)

Contact form

The image below shows the main part of my webpage the contact form.




```
// PAGE LOAD ENTRANCE ANIMATIONS
// --------------------------------
setTimeout(() => {
  if (document.querySelector(".box")) document.querySelector(".box").classList.add("show");
  if (document.querySelector(".LLBox")) document.querySelector(".LLBox").classList.add("show");
  if (document.querySelector(".LRBox")) document.querySelector(".LRBox").classList.add("show");
}, 200);
```

```
/* MAINBODY CONTACT BOX
--------------------------- */
.box {
    max-width: 1000px;
    margin: auto;
    margin-top: 40px;
    background: #b2eaec;
    padding: 50px 50px;
    display: flex;
    gap: 50px;
    border-radius: 6px;

    opacity: 0;
    transform: translateY(40px);
    transition: all 0.8s ease;
}

/* SHOW ANIMATION */
.box.show {
    opacity: 1;
    transform: translateY(0);
}
```

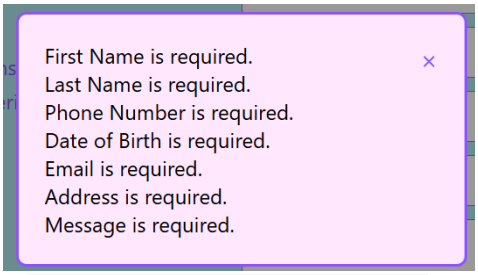
I added each section including this in a box container to make the design look clean and to separate each element. I used the colour scheme our group picked out to keep it consistent looking. I added clear labels on the form inputs using placeholder text to make is easily accessible to the user. When the webpage loads in the box has an animation where it fades in and moves up, I added this to make the website less static, and I used mainly CSS and a small amount of JavaScript to achieve this as I thought it would be the most efficient way . The JavaScript works by waiting 200 milliseconds after the page loads then checks if the box element exists and then adds the CSS Class ‘show’. The default CSS for the box element opacity is 0 so it is invisible and the transform moves the element down 40px,

so when the show CSS is added it changes the opacity to 1 and transforms the box back up to its original position. I used the same code for each of the 3 boxes on this page.

The main part of this section is the form and the form validation. The form validation was achieved using JavaScript and took me a while to get it right – specifically the error message which informs the user of any errors in the form.

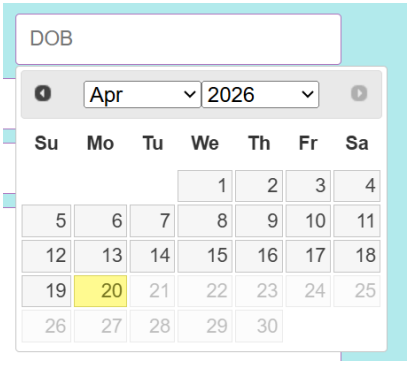
```javascript
// FORM VALIDATION FUNCTION
// ----------------------------
function validateForm() {
  let errors = [];

  const firstName = document.getElementById('firstName').value.trim();
  const lastName = document.getElementById('lastName').value.trim();
  const phone = document.getElementById('phonenum').value.trim();
  const dob = document.getElementById('dob').value.trim();
  const email = document.getElementById('email').value.trim();
  const address = document.getElementById('address').value.trim();
  const message = document.getElementById('message').value.trim();

  if (firstName === '') errors.push("First Name is required.");
  if (lastName === '') errors.push("Last Name is required.");

  if (phone === '') errors.push("Phone Number is required.");

  if (dob === '') errors.push("Date of Birth is required.");

  const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
  if (email === '') {
    errors.push("Email is required.");
  } else if (!emailPattern.test(email)) {
    errors.push("Invalid email address.");
  }

  if (address === '') errors.push("Address is required.");
  if (message === '') errors.push("Message is required.");

  if (errors.length > 0) {
    showErrors(errors);
    return false;
  }

  alert("Form submitted successfully!");
  return true;
}
```

The image to the right shows the main JavaScript function which holds the code for the validation – it runs after the submit button is clicked. I added a hover animation and a submit animation to the button to make it clear to the user that it works. This code works by creating a variable for each value entered into each form box and then uses if statement to check if it matches the requirements. Each element is required so it will check to see if the variable is empty or not, the email one also checks to ensure it has a valid email. If there is no errors then an alert pops up to inform the user the form has been submitted successfully – if there is any errors then each error message is added to a list and this list will then pop up in a box on the screen which the user can then close using the X in the corner after they have read it. As the user fixes the information in the form and resubmits the error messages will no longer appear.



I used jQuery for the DOB which is why there is not much validation in the JavaScript for it. It was easy to use jQuery as it makes it much easier for the user as they can pick a date off the calendar instead of typing it in manually – this makes the website more accessible and it is very user friendly. The jQuery also ensures that the date is not a date from the future as that would not be a valid DOB. jQuery is great as it makes some tasks a lot quicker and easier to do as I was able to do this in only a few lines, unfortunately this was the only jQuery I added to my webpage as I was unsure where else I would be able to add it or found it easier to not use it. Maybe if I had of had more time I would of found more ways

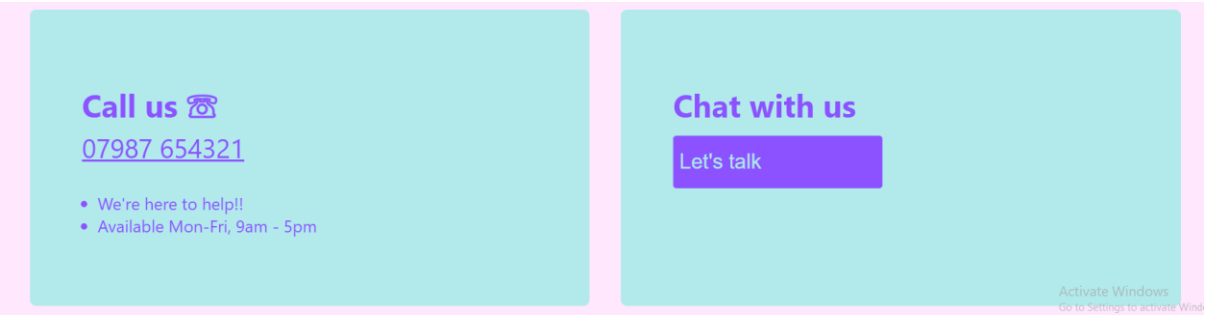
```html
<script>
    $(function () {
        $("#dob").datepicker({
            changeMonth: true,
            changeYear: true,
            yearRange: "1900:2026",
            dateFormat: "dd/mm/yy",
            maxDate: 0
        });
    });
</script>
```

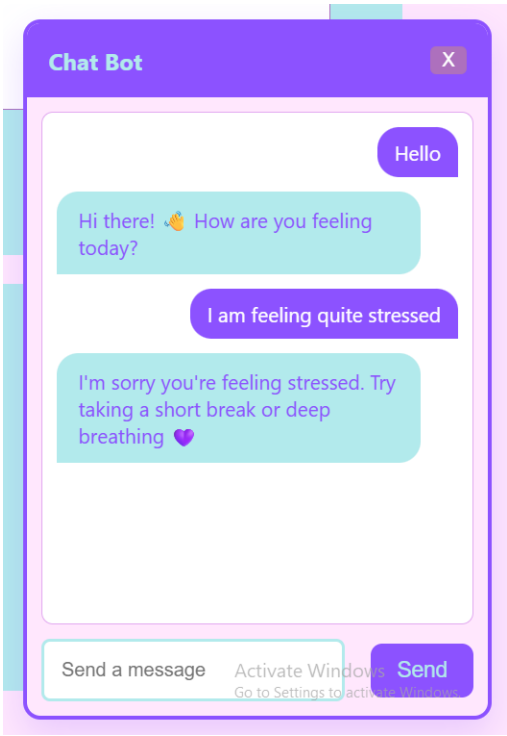


to include jQuery but I was working on other bits of coursework at the same time so my time was limited.

Chat bot

The next section of my webpage includes 2 more boxes – these were originally not part of my design but I felt like it needed more.





The first box is just a simple box with text in it. The box on the right has a button which when clicks opens up a chat bot popup – I decided to add this as when researching other websites, I found that a lot of them had one of these. I found this quite challenging to do but I do enjoy a challenge, so I didn't mind it. There were a few parts of this I struggled with and had to get a small amount of help from AI to fix things. This Chat bot works by allowing users to send a message which appear to the right and then is followed up by a pre-made message on the left, there is a few variations of messages and will change depending on certain words sent by the user - for example a message containing 'stress' will have a different message.

I used CSS to style this to match the rest of my webpage. I used a large amount of JavaScript for this section as there was multiple buttons and the response feature require a lot of JavaScript Aswell. While reflecting on this I think it would have been cool to make this box draggable so that the user can move it around the screen, but I think this might be too advanced for me to do.

Some more of the features this chat bot has is that the messages are animated when they appear and there is a slight delay when the bot message appears to simulate it thinking and writing the message itself – it is also possible to send a message by hitting the enter key Aswell as the send button as this is a common feature on other websites.

As messages are sent they will automatically scroll to the latest messages sent so as not to confuse the user where their message went, this was very easy to do as it only requires one line of code and makes a big difference to the user.

```
// show latest message
chatMessages.scrollTop = chatMessages.scrollHeight;
```

For each of the boxes I made them flexboxes using CSS to hopefully keep the layout responsive and so that it adapts to different screen sizes.

Testing

Since the contact form had a lot of validation, I had to test it multiple times with incorrect data and correct data to ensure the validation worked properly. I also tested the chat box by entering in different phrases with the keywords and without the keywords it looks for to see what the response would be and to ensure that it was the correct one.

Other pages

The page that I developed was the contact page, but the other pages included on our website were the home page, about us page, coping tools page and resource page. These were the pages we felt were crucial for our website. Each of us kept a consistent colour palette and similar design to make the whole website feel very consistent.

Conclusion

Overall we spent a lot of time and effort on our website with actual passion to have a good outcome, although there is a few things we might change we are pleased with how our website has turned out and we believe it meets each condition as it is accessible, has interactive features, has good validation, use of jQuery and consistency. We used the website w3Schools to help us code some parts of our website when there were bits, we couldn't fully do ourselves as it was very helpful.

We tested each webpage multiple times to ensure that everything worked well and we used GitHub to share our code with each other. I believe we worked very well as a group, and the good collaboration improved the quality of our website.

References

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Joseph Davidson Individual report

For this project, I made use of multiple web technologies in the About Us page, being HTML5, CSS, JavaScript and frameworks including jQuery & jQuery UI, Web storage API, localStorage and regex.

The chosen brand focus for our group is on health and wellbeing. Based on this, each element of the website was created with user calming measures and website accessibility, UX design

I created the slogan “Stress Less, Learn More”, taking to name company “StressLess”. As a group we looked online for the creation of our logo and decided on a template from Canva, which we adapted using our colour scheme to create this logo:

Stress Less, Learn more

Stress Less, Learn more

We chose this design for its clear visual storytelling & messaging that our website aims to transform mental clutter to calming clarity, shown by the overlapping squiggle lines into the StressLess brand name, ensuring a clean, minimalist design.

Contrast is created against the pink lines putting the StressLess as the solution to user’s problems. The controlled spiral on the right represents the result of the user’s journey using Stressless to improve their wellbeing.

The audience of the website was decided to be university students. I tailored the background of the company that it was made by former students, who truly understood the pressures of university life and of exams, deadlines and worries that relate to the user. “StressLess was made by students, for students”. This choice ensured to form a close connection with users.

2: Requirements

Before starting the design of our pages, each group member planned our page designs carefully to implement all of the below requirements from the specification.

- *Use of client-side technologies (HTML, CSS, JavaScript, frameworks such as jQuery)*
- *All Code should be standards-compliant and structured using best practices.*
- *Features like image slideshows, animation, scrollable areas, banners, hover effects, and real-time UI feedback should be implemented using JavaScript.*
- *Extensive form validation.*
- *Apply UX design principles.*
- *Must be capable of gathering data*
- *Use event handling techniques (for example, animations, banners, image effects, date/time, form validation, calculator, and cookies/LocalStorage*

<u>Website Design</u>

As our branding of a health & wellbeing website, we researched services that offer support and tools for users physical and mental health that we would tailor to our own website. We ensured to have calming features from the layout of the content, the colour scheme used and UI elements that are simplistic and easy to navigate for users, which was our priority for our site.

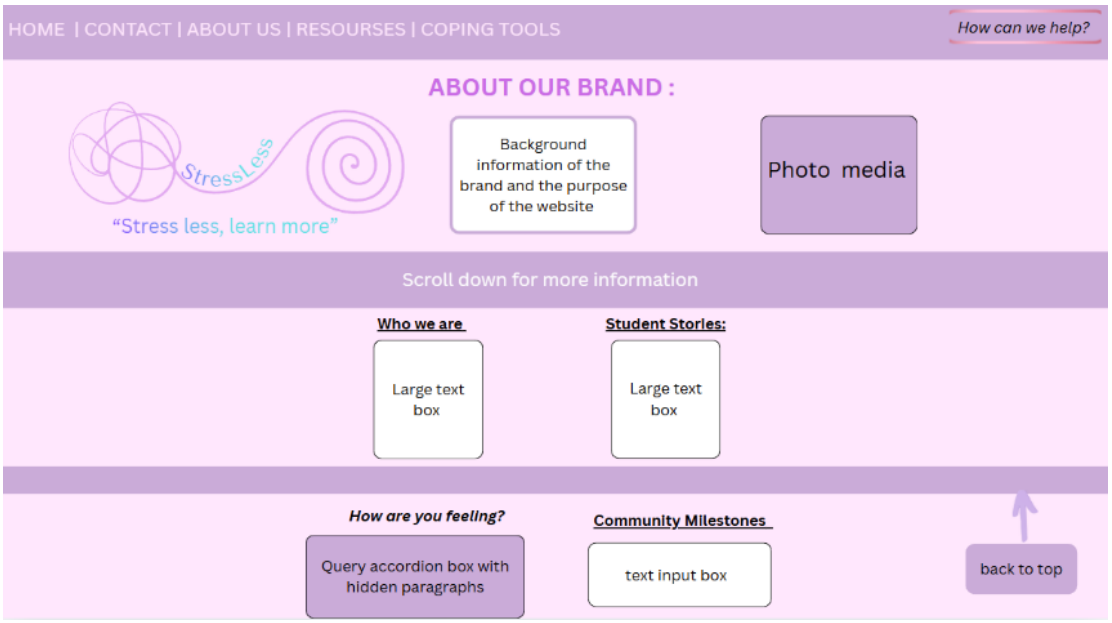
As part of our website design, we first made sure to use the same Header/footers on each page of the website so that it is easy to navigate and a familiar layout for users. We also used the same fonts across all 5 pages for to reinforce brand identity.

User experience and interactivity was incorporated in all pages through a sticky navigation bar, ensuring primary page navigation is essential at each part of the page. Hover elements were also used for containers and buttons, signifying that these elements can be engaged with to confirm user actions. A chat bot is included in each page of the website on the bottom left side, so that users can instantly have interaction for answers. Clicking the chat circle, typing the word “stress” and hitting the send button prompts a tailored response to help the user’s wellbeing.

For the colour scheme, it implemented across all pages of our website, chosen as high contrast against text and images, using Electric Aqua and Medium slate blue. I additionally used “Rebecca purple” for the page dividers to create clear visual hierarchy of the page and its sections.

Below is my wireframe design for the About us page, which I used to visualize the requirements

Wireframe:



I created the slogan “Stress Less, Learn More”, which we took to name the company “StressLess”. As a group we looked online for the creation of our logo and decided on a template from Canva, which we adapted using our colour scheme to create this logo:

We chose this design for its visual storytelling that our website aims to transform mental clutter to calming clarity, shown by the overlapping squiggle lines into the StressLess brand name, ensuring a minimalist design.

This helps to create contrast and put StressLess as the solution to user’s mental health problems. The controlled spiral on the right represents the result of the user’s journey using Stressless to improve their wellbeing.

As a group, we decided on the final navigation bar which includes are logo, button links, my search bar and contact form using our cyan colour, ensuring users are drawn toward each element.

Search Bar:

For my About-us page, I implemented a search bar so that users can easily find information they are looking for as quickly as possible when they enter the website.

I had trouble in choosing how I wanted the search to look against the rest of the elements, deciding on a high-contrast white background to distinct against the purple navigation bar, to show that the search bar is interactive. On hover, the purple icon provides immediate visual feedback that the element is clickable, reducing user anxiety.

In the HTML, I used the <mark> tag to provide visual confirmation that users found what they were looking for. Using jQuery for the coding of the search bar, usability was achieved through .animate() and scrollTop, which positions the screen directly in front of the highlighted text users were looking for. I ensured constraint handling by requiring term.length > 2, so that one letter words don't get picked up. I used a negative lookahead regular expression to ensure HTML tags and part of image sources are not highlighted, only the intended text.

Screenshot of the jQuery code of the search bar:

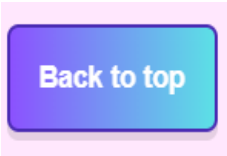
```jquery
$('.navbar__search form').on('submit', function (e) {
    e.preventDefault();

    const term = $(this).find('.search__input').val().toLowerCase();
    const content = $('.middle-content, .bottom-section-container');

    $('mark').contents().unwrap();

    if (term.length > 2) {
        content.each(function () {
            const instance = $(this);
            const html = instance.html();

            const regex = new RegExp("(?![^<]*>)" + term, "ig");
            const newHtml = html.replace(regex, (match) => `<mark>${match}</mark>`);

            instance.html(newHtml);
        });

        if ($('mark').length > 0) {
            $('html, body').animate({
                scrollTop: $("mark").first().offset().top - 120
            }, 500);
        }
    }
});
```

On the page:

On hover:

Navigation button paragraph:

To enable seamless navigation across each page, I created the "Back to top" navigation button using an event listener that only appears visible at a 300-pixel threshold when the user scrolls down, minimizing visual clutter. I ensured the button works across different browsers e.g. chrome, Firefox by checking both document.body.scrollTop and document.documentElement.scrollTop.



To enhance interactivity, I placed the button using position: fixed; at the bottom-right of the page, following website accessibility standards. The UX was improved by enabling behaviour: ‘smooth’ to make an intuitive transition to the top of the page.

Screenshot of the JavaScript

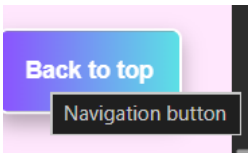
```javascript
window.onscroll = function () {
    scrollFunction();
};

function scrollFunction() {
    let navbutton = document.getElementById("navBtn");

    if (document.body.scrollTop > 300 || document.documentElement.scrollTop > 300) {
        navbutton.style.display = "block";
    } else {
        navbutton.style.display = "none";
    }
}

function topFunction() {
    window.scrollTo({ top: 0, behavior: 'smooth' });
}
```

I used CSS in creating the gradient background of button, with the text “Back to top” displayed clearly for enhanced navigation for the user. On hover, a title attribute appears and the background changes from a purple border to white, confirming interactivity with this element. I included an upward movement animation using translateY(-3px), moving the button 3 pixels up. On click, the active state changes and moves 1 pixel down, mimicking the button being pressed.

Screenshot of the CSS navigation button

```css
#navBtn {
    display: none;
    background: linear-gradient(90deg, #8c52ff, #5ce1e6);
    position: fixed;
    bottom: 30px;
    right: 30px;
    z-index: 99;
    font-size: 14px;
    font-weight: bold;
    border: 2px solid #4b2cb3;
    color: white;
    cursor: pointer;
    padding: 15px;
    border-radius: 6px;
    box-shadow: 0 4px 0px rgba(0,0,0,0.1);
    transition: all 0.3s ease;
}


/* on hover, background changes */
#navBtn:hover {
    background-color: #000000;
    border-color: #ffffff;
    transform: translateY(-3px);
    box-shadow: 0 6px 12px rgba(0,0,0,0.3);
    color: #ffffff;
}

#navBtn:active {
    transform: translateY(-1px);
}
```

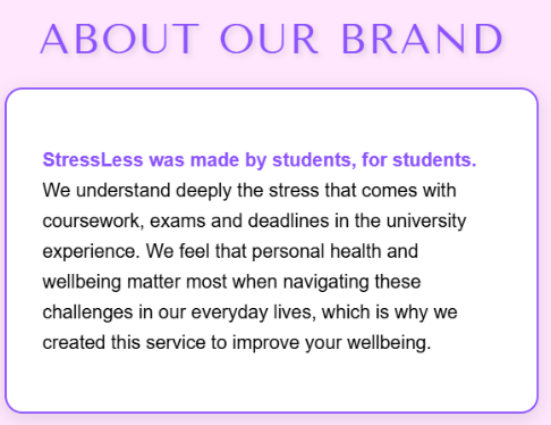
On hover



Text box paragraph:

When the user enters the site, the “About our brand” top text box grabs the user’s attention using animations including @keyframes and fadeInUp . When the page is opened, the helps to create a more dynamic user experience. The text box smoothly fades up the page using opacity and transform, establishing a professional look when the page is opened.

```
@keyframes fadeInUp {
    from {
        opacity: 0;
        transform: translateY(30px);
    }
    to {
        opacity: 1;
        transform: translateY(0);
    }
}
```



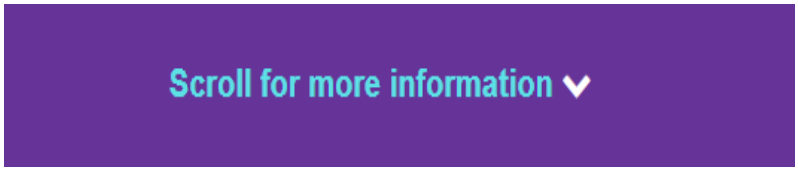
Inside the hero section on the right, I added a student support interaction photo to provide visual context of our services. The image helps to break up text, making the content easier to digest and more visually appealing. I styled the image with a 15px border and used margin-top: 60px; to so elements aren’t cluttered, being easy to navigate.

For the text of the page, I globally styled the <b> tag HTML text to the brand’s medium slate blue across the text of the page and added font-weight to the CSS. This helped to improve readability, so vital keywords are picked up with ease.

```
b, strong {
    color: #8c52ff;
    font-weight: 700;
}
```

```
.student-image-column {
    grid-column: auto;
    display: flex;
    justify-content: flex-start;
    width: 100%;
}

.student-support {
    width: 100%;
    max-width: 400px;
    height: auto;
    margin-top: 60px;
    border-radius: 15px;
    border: 4px solid rebeccapurple;
    box-shadow: 0 4px 15px rgba(0,0,0,0.1);
    display: block;
}
```

The top page divider was used to create clear visual hierarchy of the page, using the background-colour Rebeccapurple to add contrast. The “Scroll down for more information” call-to-action text ensures the user is informed at each step of the website. On hover, the colour to #5ce1e6 appears to prompt the user to scroll. Beside it, I placed arrow using the Font Awesome Library, which was very useful when adding icons to my page.




```html
<div class="page-divider">
    <div class="scroll-prompt">
        <span>Scroll for more information </span>
        <i class="fa fa-chevron-down"></i>
    </div>
</div>
```

Who we are & Student Stories:

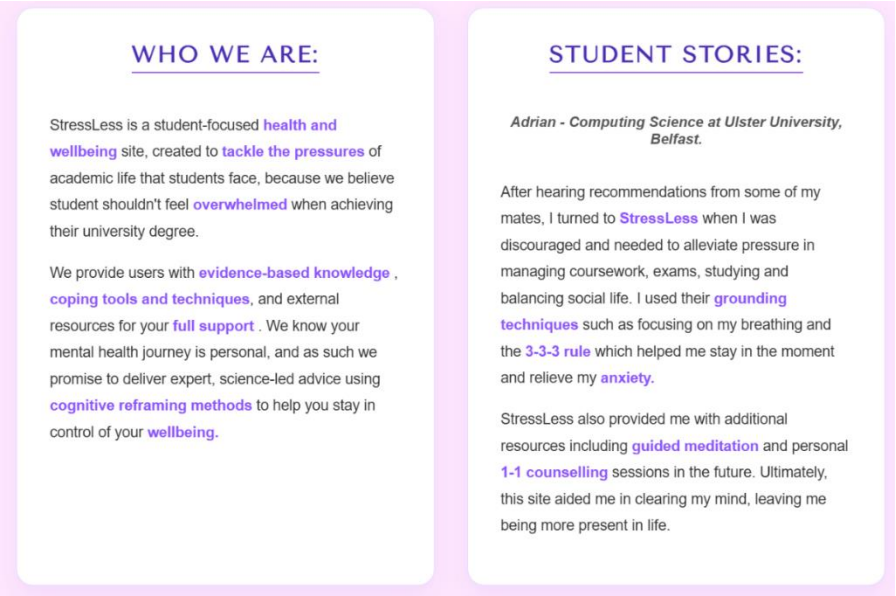
To improve UI feedback, I added a scroll-triggered reveal to the middle section of the page to ensure elements to help improve the user experience as they navigate down the page. Upon entering the page, only the top hero section is visible to the user, ensuring elements are not visually cluttered to reduce cognitive load. For this I used opacity: 0; and transform: translate(30px); and reveal-active to change the opacity and position when the scrolling action is complete.

```css
.who-we-are-container, .stories-container {
    display: flex;
    flex-direction: column;
    justify-content: flex-start;
    align-items: flex-start;
    gap: 10px;

    background: #ffffff;
    padding: 40px;
    border-radius: 20px;
    box-shadow: 0 10px 30px rgba(140, 82, 255, 0.1);
    border: 1px solid rgba(140, 82, 255, 0.2);

    opacity: 0;
    transform: translateY(30px);
    transition: all 0.8s ease-out;
}

.reveal-active {
    opacity: 1;
    transform: translateY(0);
}
```

For the titles of the containers, I used Tenor Sans with text underlined. I had trouble with visualizing how I wanted it to look but settled on a CSS grid layout to create a balanced visual hierarchy between the brand information and student stories. I increased interactivity by adding transition-delay: 0.2s, to this container to create a staggered reveal effect, guiding the user to read the “who we are” container first. Whitespace is provided on each side of the containers, ensuring users are focused on those two elements.

```css
.stories-container {
    display: flex;
    flex-direction: column;
    justify-content: flex-start;
    align-items: flex-start;
    height: 100%;

    transition-delay: 0.2s;
}
```

Screenshot of the containers



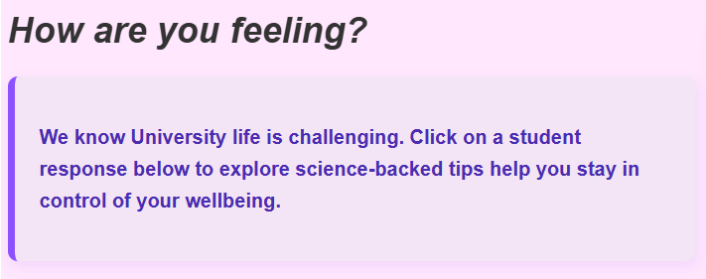
Student quote bar:

I implemented a student testimonial container to act as a visual anchor between the 'Who we are' and 'Student Stories' sections. Using a blue background, I created a calming contrast against vibrant brand colours across the page, enabling a focused environment on the quote itself. I utilized a Flexbox wrapper to define the max width and margin to be centred and responsive. For mobile optimisation, I included an @media query to maintain a high level of readability, so the quote doesn't become cramped.

Additionally, I used a quote icon left and right from the Font awesome library, ensuring users know this is a direct quote from a student using the website. When users see that a specific named person, "Adrian" had good experience using StressLess, they are more likely to interact with the site.

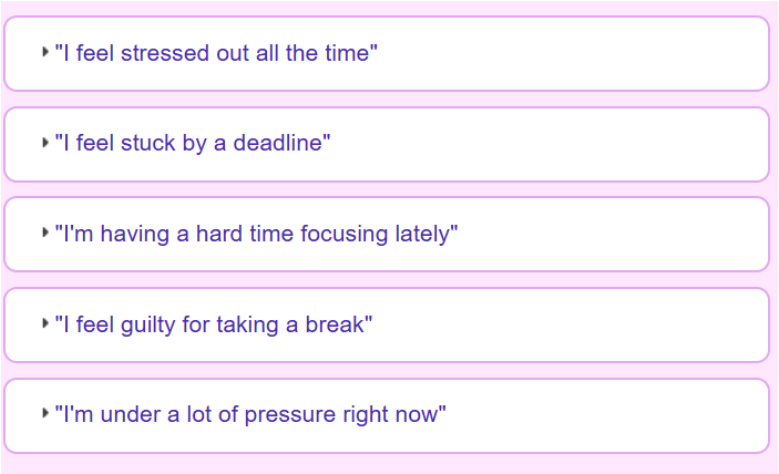
Accordion:

I implemented an interactive accordion section at the bottom section, providing wellbeing tips commonly asked student quotes to help improve their mental and physical health. Direct call-to-action is



given, so the user know that the accordion can be interacted with.

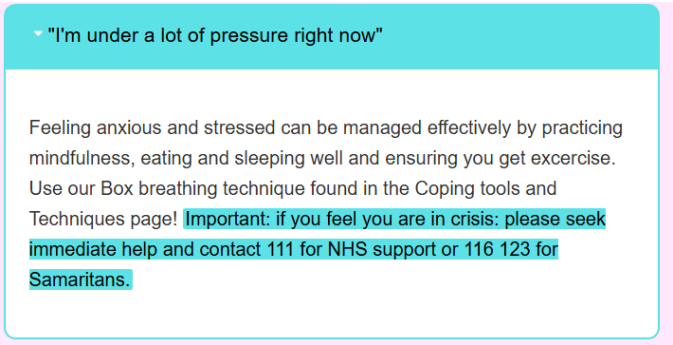
Screenshot of accordion quotes



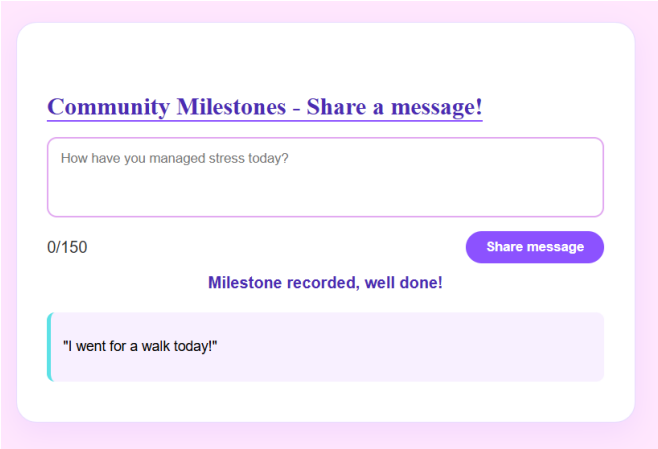
Using CSS, I added an On-hover class to each quote, changing the border cyan to encourage engagement with the accordion. Using progressive disclosure UX design, I added an On-click reveal to a StressLess curated response, so users can select the advice most relevant to their mood.

```
#accordion-questions h4 {
    background-color: #ffffff;
    color: #4b2cb3;
    padding: 15px 20px;
    margin-top: 10px;
    border: 2px solid #e2a9f1;
    border-radius: 10px;
    cursor: pointer;
    font-size: 17px;
    transition: all 0.3s ease;
    display: flex;
    align-items: center;
    outline: none;
}

#accordion-questions h4:hover {
    background-color: #f0fdff;
    border-color: #5ce1e6;
    color: #333;
}

#accordion-questions h4.ui-state-active {
    background-color: #5ce1e6;
    color: #000000;
    border-color: #5ce1e6;
    border-bottom-left-radius: 0;
    border-bottom-right-radius: 0;
    margin-bottom: 0;
}
```

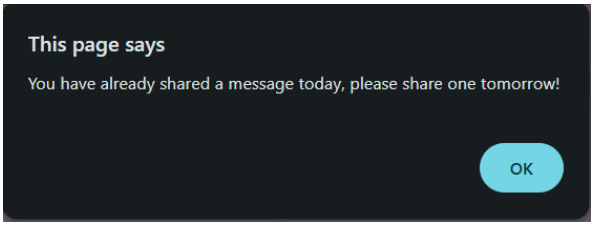
Under the “I’m under a lot of pressure right now” accordion response, I included a high-visibilty <mark> tag the NHS and Samaritans helplines for immediate support for users under pressure. Text is highlighted to show importance.



The Community Milestones feature allows users to anonymously share wellbeing success wins, creating a sense of community success using StressLess. I implemented an input box that utilizes LocalStorage to save messages underneath. Real-time feedback is given by the text: “Milestone recorded, well done!” in purple.



I created a 48-hour message deletion logic using parseInt, to keeping feed clear of clutter, allowing other users of the site to write their message. Additionally, a limit was made that only one message can be sent per day.

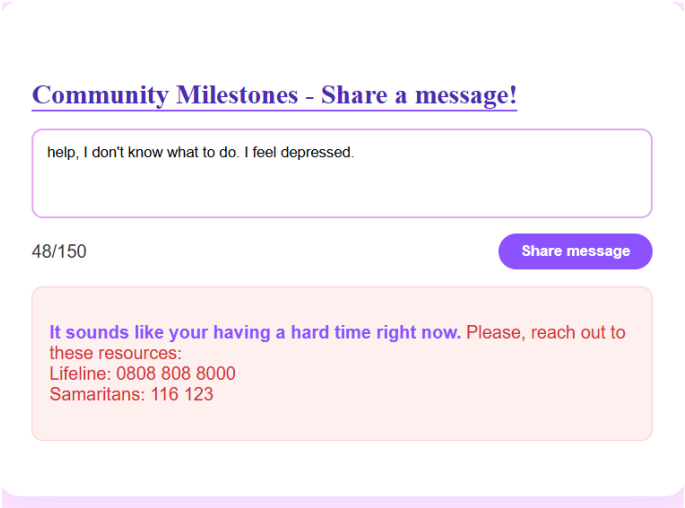



```
$('#milestone-btn').on('click', function () {

    const lastSubmit = parseInt(localStorage.getItem('user_submitted_message'));
    const now = Date.now();
    const twentyFourHours = 24 * 60 * 60 * 1000;
```

Using jQuery, I added focus() and blur() that changes the border colour to #8c52ff when the user clicks the text area, signalling to the user that it is ready to receive input.

```
$('#milestone-input')
    .on('focus', function () {
        $('.milestone-wrapper').addClass('focused');
    })
    .on('blur', function () {
        $('.milestone-wrapper').removeClass('focused');
    });
```

I also used keyword filtering through the crisis Words array. Key words like “depressed” & “suicide” are scanned in the algorithm and emergency resource links are provided to provide professional help immediately.



Screenshot to show crisis filters

```
const crisisWords = ["help", "hurt", "die", "suicide", "depressed", "emergency", "hopeless"];

let savedMessages = JSON.parse(localStorage.getItem('community_messages')) || [];
savedMessages.forEach(msg => appendMessageToFeed(msg));
```

Finally, I imposed a 200-word limit to the text box highlighted in red, so that user’s do not go over the limit and take up the page.

Reflection:

This project strengthened my understanding on UX principles, use of jQuery, CSS and the importance of valuing the user’s experience of using the website. If I were to redo, I would improve keyboard navigation and adding ARIA labels to my HTML, further enhancing accessibility.

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COM109 Coursework 2 – B00950166

MALAK ELABBASY

1. Introduction

The development of the **Stress Less** resources webpage was driven by both technical requirements and a strong emphasis on user-centred design. Before beginning implementation, I conducted a detailed review of the project specification to identify all functional and non-functional requirements. This ensured the final product would meet technical criteria while providing a meaningful, accessible experience for users.

Unlike typical web development projects, this website addresses a sensitive topic—stress management—requiring careful consideration of usability, accessibility, and emotional design. My approach combined technical implementation with psychological awareness, ensuring the website is not overwhelming, visually aggressive, or difficult to navigate.

The webpage was developed using **HTML5**, **CSS3**, **JavaScript**, and the **jQuery** library, with each technology playing a distinct role: HTML provides semantic structure, CSS ensures consistent and accessible styling, and JavaScript (with jQuery) enables dynamic interaction and real-time feedback.

2. Requirement Identification and Implementation

Following analysis of the project brief, several key requirements were identified:

- Use of client-side technologies (HTML, CSS, JavaScript, jQuery)
- Standards-compliant, well-structured code
- Implementation of dynamic and interactive features
- Application of UX/UI design principles
- Use of event-driven programming
- Inclusion of engaging and meaningful content

Rather than simply implementing these features, I ensured each requirement was fulfilled with clear intent and justification.

2.1 Client-Side Technologies

The website demonstrates comprehensive use of client-side technologies:

- **HTML5** structures content using semantic elements such as <nav>, <header>, and <footer>, improving accessibility and maintainability.

- **CSS3** creates a consistent design system across the page, including layout, typography, colour schemes, and responsive behaviour.
- **JavaScript and jQuery** implement interactivity, including accordions, event handling, and the mini-game.

For example, jQuery initialises accordion components efficiently:

```
$("#mainAccordion").accordion({
    collapsible: true,
    heightStyle: "content"
});

$("#faqAccordion").accordion({
    collapsible: true,
    heightStyle: "content"
});
```

This demonstrates effective use of a framework to enhance functionality without unnecessary complexity.

2.2 Code Structure and Best Practices

The project follows best practices through a clear separation of concerns:

- **Structure** (HTML)
- **Styling** (CSS)
- **Behaviour** (JavaScript)

This modular approach ensures scalability and maintainability. Reusable CSS classes (e.g., .navbar__container, .footer__links) avoid redundancy and improve consistency. External stylesheets and scripts allow global updates without modifying individual elements, demonstrating efficient coding practice.

2.3 Event Handling and Interactivity

Event-driven programming plays a central role in the website's functionality:

- **Click events** (mini-game buttons)
- **Scroll events** (navigation button visibility)
- **UI interactions** (accordion expansion)

The scroll event enhances usability by dynamically responding to user behaviour:

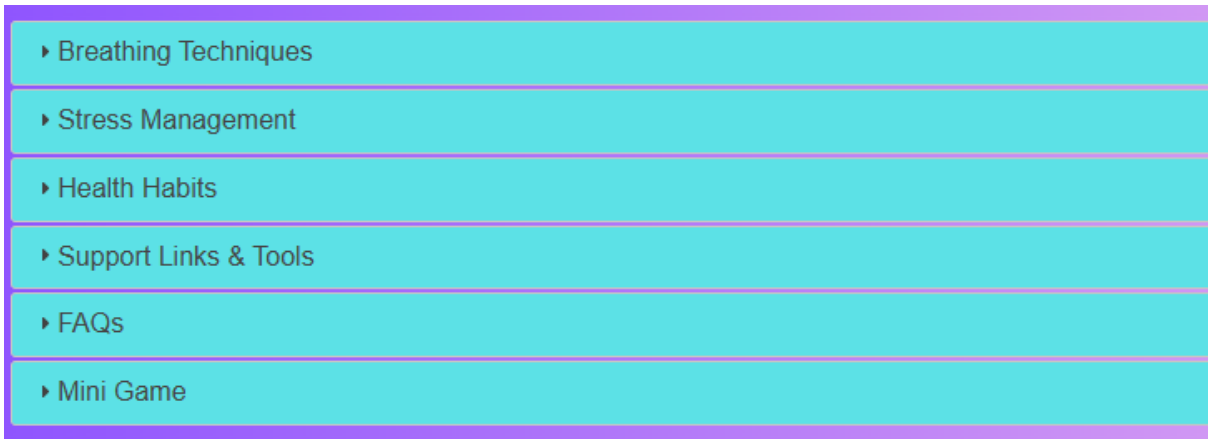
```javascript
window.onscroll = function () {
    let navbutton = document.getElementById("navBtn");

    if (!navbutton) return; // prevents errors

    if (document.body.scrollTop > 300 || document.documentElement.scrollTop > 300) {
        navbutton.style.display = "block";
    } else {
        navbutton.style.display = "none";
    }
};
```

This implementation uses the **Browser Object Model (BOM)** to access the window's scroll position and the **Document Object Model (DOM)** to manipulate the button's display property, creating a seamless user experience where navigation aids appear precisely when needed.

3. UX/UI Design Principles

A significant portion of this project focused on UX and UI design, particularly in preventing cognitive overload and promoting calm interaction.

3.1 Preventing Information Overload Through Progressive Disclosure



One of the most critical design decisions was implementing accordion components using jQuery UI. Rather than displaying all information simultaneously, content is segmented into expandable sections. This approach is rooted in cognitive load theory, first proposed by John Sweller, which states that the human brain has a limited capacity for processing information at any one time. When too much information is presented simultaneously, working memory becomes overloaded, leading to confusion, disengagement, and in the context of this website, potentially increased anxiety in users who are already struggling.

For a stress management platform, this risk is amplified. A user arriving at the page in a heightened state of stress or anxiety is far more vulnerable to feeling overwhelmed by dense, unstructured content. By hiding content behind expandable headers, the

accordion system ensures users are only ever processing one topic at a time, significantly reducing this risk.

Justification:

- Reduces cognitive load by presenting information in digestible chunks
- Grants users control over their experience, fostering autonomy
- Improves readability and focus by eliminating visual clutter
- Prevents overwhelming users—particularly critical for stress-related content

The nested accordion structure in the FAQ section exemplifies progressive disclosure at multiple levels: users first see high-level categories, then expand to reveal specific questions, and finally access detailed answers. This hierarchical approach mirrors how users naturally seek information, reducing anxiety about navigating complex content. It also respects user autonomy, allowing individuals to engage only with topics relevant to their personal situation rather than being forced through irrelevant material.

The collapsible: true parameter is deliberately chosen to allow all sections to be closed simultaneously, giving users complete control rather than forcing one section to remain open. This small but meaningful decision reinforces the website's core philosophy: the user is always in control of their experience.

.

```
$("#faqAccordion").accordion({
    collapsible: true,
    heightStyle: "content"
});
```

The collapsible: true parameter is deliberately chosen to allow all sections to be closed simultaneously, giving users complete control rather than forcing one section to remain open.

3.2 Colour Scheme and Emotional Design

The colour palette was carefully selected based on colour psychology principles:

background: linear-gradient(to right, #8c52ff, #e2a9f1, #5ce1e6);

Rationale:

Colour	Hex Code	Psychological Association
Purple	#8c52ff	Calmness, creativity, introspection

Colour	Hex Code	Psychological Association
Pink	#e2a9f1	Nurturing, compassion, emotional safety
Cyan	#5ce1e6	Tranquillity, clarity, refreshment

Design Justification:

- **Soft gradients** create a calming visual effect, avoiding harsh transitions that could increase stress
- **Pastel tones** are associated with relaxation and clarity, deliberately avoiding high-saturation colours that may trigger anxiety
- The **gradient direction** (left to right) follows natural reading patterns in Western cultures, creating subconscious visual harmony
- **Contrast ratios** between text and backgrounds exceed WCAG 2.1 AA standards, ensuring legibility for users with visual impairments

3.3 Typography and Accessibility

The typography system prioritises accessibility and readability:

```
font-family: Arial, sans-serif;
background: linear-gradient(to right, #8c52ff, #e2a9f1, #5ce1e6);
```

Sans-serif fonts were deliberately chosen for several accessibility reasons:

- **Dyslexia-friendly**: Sans-serif fonts reduce visual confusion caused by serifs, which can cause letters to appear to merge or flip for dyslexic readers
- **Screen optimisation**: Digital screens render sans-serif fonts more clearly at various sizes, reducing eye strain
- **Cognitive load reduction**: Clean letterforms require less processing effort, particularly important for users experiencing stress or anxiety

Line height and spacing were calculated to enhance readability:

```
p, li {
    margin-top: 10px;
    line-height: 1.7;
}
```

A line height of 1.7 exceeds the minimum 1.5 recommended by WCAG guidelines, providing additional breathing room that reduces the cramped appearance common on

text-heavy pages. This spacing is particularly beneficial for users with attention difficulties or visual processing challenges.

4. Interactive Features: Technical and Conceptual Analysis

4.1 Mini-Game Implementation

The mini-game represents a deliberate therapeutic intervention disguised as simple entertainment. The implementation uses event-driven programming with precise timing control:

```
$("#startGame").click(function () {
    score = 0;
    active = true;
    $("#score").text("Score: 0");
    $("#clickBtn").prop("disabled", false);

    setTimeout(() => {
        active = false;
        $("#clickBtn").prop("disabled", true);
        alert("Time's up! Score: " + score);
    }, 10000);
});

$("#clickBtn").click(function () {
    if (active) {
        score++;
        $("#score").text("Score: " + score);
    }
});
```

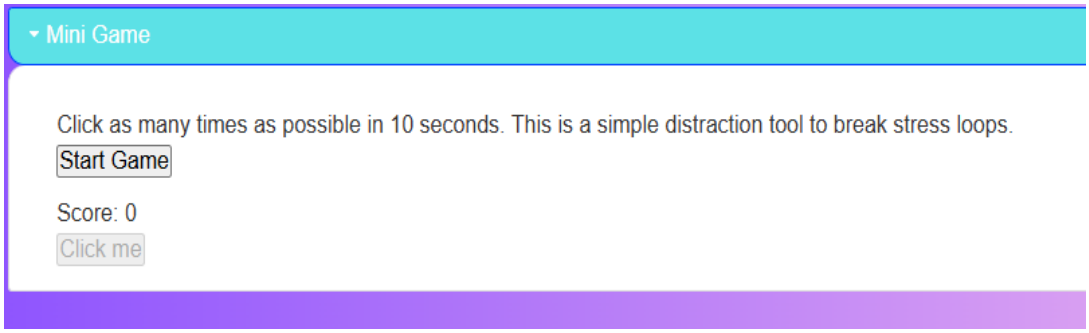
Technical Analysis:

- **State management:** The active boolean variable controls game state, preventing score manipulation outside the timed window
- **DOM manipulation:** jQuery's .text() method provides real-time feedback, updating the score display instantly
- **Timer control:** setTimeout() creates a precise 10-second window, using the BOM's timing capabilities
- **Button state management:** .prop("disabled", false/true) prevents user interaction outside valid game states, ensuring predictable behaviour

Psychological and Therapeutic Justification:

The game was intentionally designed with specific therapeutic properties grounded in established psychological research:

1. **Simplicity:** The single-action mechanic requires minimal cognitive processing, making it accessible during high-stress states when complex decision-making becomes difficult. Research into stress-induced cognitive narrowing shows that anxiety reduces a person's capacity for multi-step thinking, meaning simple, single-focus tasks are far more effective as interventions than complex ones.
2. **Repetitive motion:** Rhythmic clicking creates a grounding effect similar to techniques used in anxiety management, such as tapping or bilateral stimulation used in EMDR (Eye Movement Desensitisation and Reprocessing) therapy. Bilateral and repetitive physical stimulation has been shown to engage the parasympathetic nervous system, actively countering the fight-or-flight response triggered by stress.
3. **Short duration:** The 10-second timeframe is deliberately brief—long enough to shift attention away from stressors but short enough to prevent new anxiety about performance or time commitment. This respects the psychological principle that brief, achievable tasks build self-efficacy, a person's belief in their own ability to cope, which is directly linked to stress resilience.
4. **Immediate feedback:** The real-time score counter provides instant gratification and a measurable sense of accomplishment. This triggers a small dopamine release, a neurotransmitter associated with reward and motivation, which can temporarily elevate mood and break a negative stress cycle.
5. **Clear boundaries:** Defined start and end points create psychological safety. Users know exactly what to expect and when the task will conclude, removing uncertainty, which is itself a primary driver of anxiety. The disabled button state outside of active gameplay reinforces these boundaries at a technical level, ensuring the experience remains predictable and controlled.



4.2 Back-to-Top Navigation Button

The navigation button demonstrates sophisticated UX thinking combined with technical implementation:

```javascript
// 👇 KEEP THIS OUTSIDE the jQuery block
window.onscroll = function () {
    let navbutton = document.getElementById("navBtn");

    if (!navbutton) return; // prevents errors

    if (document.body.scrollTop > 300 || document.documentElement.scrollTop > 300) {
        navbutton.style.display = "block";
    } else {
        navbutton.style.display = "none";
    }
};

function topFunction() {
    window.scrollTo({ top: 0, behavior: 'smooth' });
}
```

Technical Features:

- **Cross-browser compatibility**: The condition checks both document.body.scrollTop and document.documentElement.scrollTop to ensure functionality across different browsers
- **Null check**: if (!navbutton) return prevents JavaScript errors if the element is missing
- **Smooth scrolling**: The behavior: 'smooth' parameter creates an animated transition rather than jarring instant movement

UX Justification:

- **Contextual appearance**: The button only appears after scrolling 300 pixels, preventing visual clutter when unnecessary
- **Reduced physical effort**: Long pages require significant scrolling; the button eliminates this repetitive strain
- **Visual feedback**: CSS transitions provide hover effects that confirm interactivity

5. External Resources and Support Links

The inclusion of external mental health resources was a deliberate design decision that significantly enhances the website's value proposition.

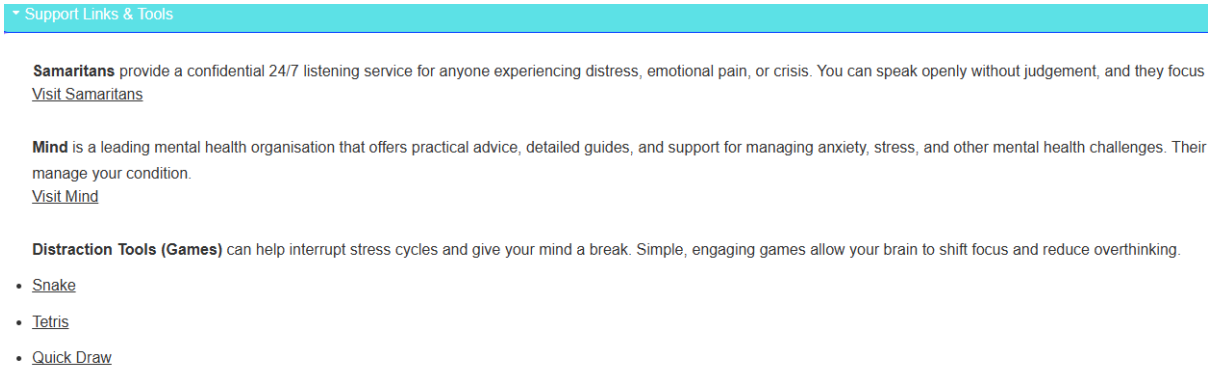
5.1 Professional Support Services

Links to **Samaritans** and **Mind** serve multiple purposes:

- **Extended support**: The website acknowledges its limitations as a self-help resource, directing users to professional services when needed
- **Credibility**: Association with established organisations increases user trust
- **Ethical responsibility**: For sensitive topics like mental health, providing professional referrals is essential

5.2 Distraction Tools

The external game links (Snake, Tetris, Quick Draw) complement the internal mini-game:



- **Variety:** Different game types suit different preferences and stress responses. Snake demands sustained focus, Tetris requires pattern recognition, and Quick Draw engages creativity — meaning users with different coping styles are more likely to find something that works for them.
- **Extended engagement:** External games offer longer distraction periods when needed. While the internal mini-game provides a brief 10-second intervention, some users may require more sustained relief to fully break a stress cycle, making these external options a valuable extension of the toolkit.
- **Evidence-based selection:** Research suggests puzzle games like Tetris can reduce intrusive thoughts and anxiety symptoms. This is because Tetris occupies the brain's visuospatial processing system — the same system responsible for generating anxious mental imagery — effectively crowding out negative thought patterns and providing targeted rather than simply general distraction.

6. Responsive Design and Mobile Accessibility

The CSS implements responsive behaviour through media queries:

```
@media screen and (max-width: 960px) {
    .navbar__container {
        display: flex;
        justify-content: space-between;
        height: 80px;
        z-index: 1;
        width: 100%;
        max-width: 1200px;
        padding: 0;
    }
```

```
@media screen and (max-width: 960px) {
  .navbar__menu {
    display: grid;
    grid-template-columns: auto;
    position: absolute;
    top: -1000px;
    opacity: 0;
    transition: all 0.5s ease;
  }
}
```

The hamburger menu transformation ensures navigation remains accessible on mobile devices, where a significant portion of users experiencing acute stress may access resources. The smooth transition animations maintain the calming aesthetic even during interface changes.

7. Conclusion

The Stress Less resources webpage successfully meets all specified requirements while demonstrating strong application of UX/UI design principles. The integration of interactive elements, accessible design, and thoughtful content organisation results in a highly functional and user-friendly website.

The project highlights the importance of balancing technical implementation with user needs, particularly when designing for sensitive topics such as mental health. Every design decision—from colour selection to game mechanics to typography—was informed by both technical best practices and psychological understanding of the target audience.

The combination of progressive disclosure through accordions, therapeutic micro-interactions through the mini-game, and comprehensive external resource integration creates a holistic stress management platform that extends beyond simple information delivery to actively support user wellbeing.

8. References

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